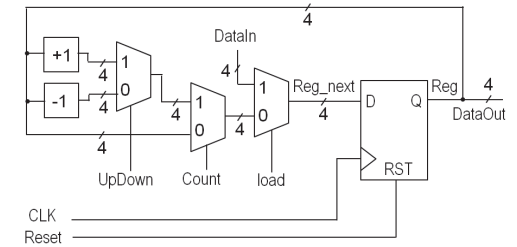


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```
library IEEE;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;
```

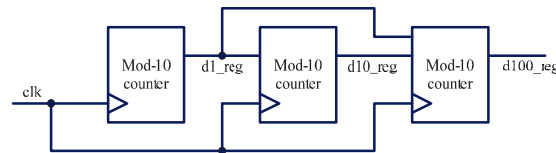
architecture two-seg of counter is
signal Reg, Reg_Next : std_logic_vector (3 downto 0);



- 1、先画flip-flop
- 2、优先级：load > count > updown

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- A decimal counter circulates the patterns in binary-coded decimal (BCD) format.
- The BCD code use 4 bits to represent a decimal number.



```
library IEEE;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;
```

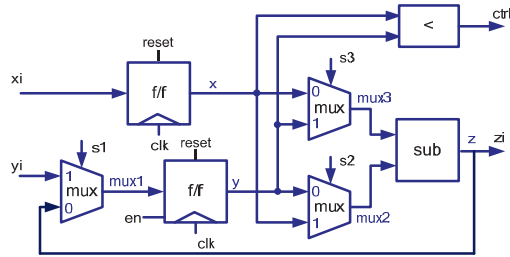
begin

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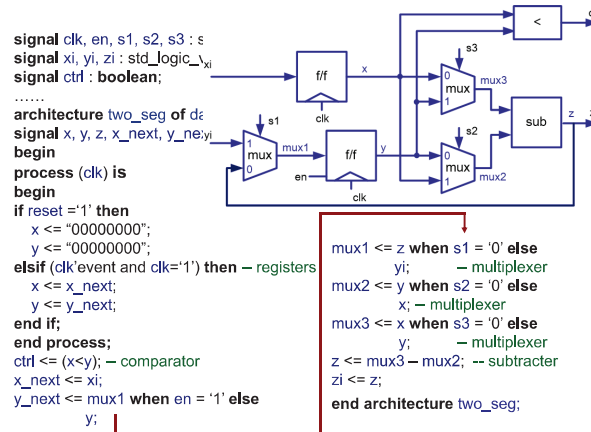
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6.3 Netlist of RTL components

- A data path usually consists of a netlist of RTL components such as function units, multiplexers, comparators, registers, etc.



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6.4 Test benches for sequential system

- All synchronous system require a system clock signal.
- A reset signal is required. The reset signal is asserted at power on to place the sequential system in its initial state.

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6.4.1 Generating a system clock

- 50% duty cycle clock

```
clock_gen: process
constant period : time := 100 ns;
begin
clk <= '0';
wait for period/2;
clk <= '1';
wait for period/2;
end process;
```

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6.4.2 Generating the system reset

- The reset signal typically
 - starts in its asserted state at power on,
 - remains in that state for a specified period of time, then
 - changes to its unasserted state, and
 - remains there for as long as power continues to be applied to the system.

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- The duration of the assertion of the reset signal is specified as
 - either a fixed time

```
reset <= '1', '0' after 160 ns;
```
 - or some multiple of the clock's period and is synchronized to the system clock

```
reset_process : process
begin
reset <= '1';
for i in 1 to 2 loop
wait until clk = '1';
end loop;
reset <= '0';
wait;
end process;
```

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6.4.3 Synchronizing stimulus generation and monitoring

```
monitor : process
constant n : integer := 1000;
variable number : integer range 0 to 999 := 0;
begin
-- 仿真写testbench时可以用variable
wait until reset <= '0';
wait for 1 ns;
for i in 0 to n loop
number := to_integer(unsigned(d100))*100+to_integer(unsigned(d10))*10
+ to_integer(unsigned(d1));
assert number = i mod n integer'image(); 将整数转换成字符串
report "count of " & integer'image(i mod n) & " failed"
severity error;
wait until clk = '1';
wait for 1 ns;
end loop;
wait;
end process;
```

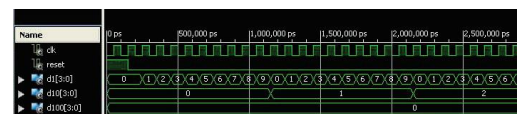
```
-- register
process (CLK, RESET) is
begin
if RESET = '1' then
d1_reg <= "0000";
d10_reg <= "0000";
d100_reg <= "0000";
elsif CLK'event and CLK='1' then
d1_reg <= d1_next;
d10_reg <= d10_next;
d100_reg <= d100_next;
end if;
end process;
```

```
-- next-state logic
d1_next <= "0000" when d1_reg = 9 else d1_reg+1;
d10_next <= "0000" when (d1_reg = 9 and d10_reg = 9) else
d10_reg+1 when d1_reg = 9 else d10_reg;
d100_next <= "0000" when (d1_reg=9 and d10_reg=9 and d100_reg=9) else
d100_reg+1 when (d1_reg=9 and d10_reg=9) else d100_reg;

-- Output logic
d1 <= d1_reg; d10 <= d10_reg; d100 <= d100_reg;
end architecture concurrent_arch;
```

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Waveforms of clk and reset



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```
use ieee.numeric_std.all;
signal x : std_logic_vector(3 downto 0); -- vector with element std_logic
signal y : unsigned(3 downto 0); -- vector with element std_logic
signal z : integer range 0 to 15;

conversion between std_logic_vector, signed, unsigned
x <= y; -- illegal assignment, type conflict
y <= x; -- illegal assignment, type conflict

x <= std_logic_vector(y); -- legal assignment
y <= unsigned(x); -- legal assignment

conversion between signed, unsigned, integer
z <= to_integer(y); -- legal assignment
z <= to_integer(unsigned(x)); -- legal assignment
y <= to_unsigned(z, 4); -- legal assignment
x <= std_logic_vector(to_unsigned(z, 4)); -- legal assignment
```

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